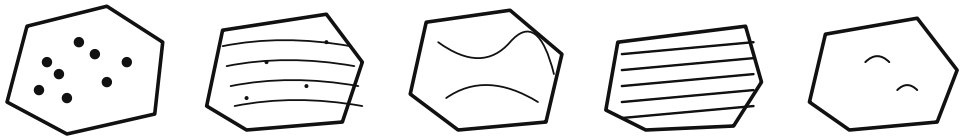
EARTH AND SPACE

Rocks, Minerals and Soils

Types of rock and their uses, minerals, how rocks break down into soil, and what soil is made of



granite sandstone marble slate chalk

In this pack

Learn it: Rocks, minerals and soils ★☆☆
Fact check: Rocks and their uses ★☆☆
Think it through: Rocks change ★★☆
Investigate: Which soil holds most water? ★★☆
Real-world case study: The brick kiln ★★★
Review and challenge: Show what you know ★★★

This pack belongs to: _____ **Class:** _____

© InquiryMind · Learn · Play · Wonder · Free resources for everyone, bringing quality and equity to education and helping teachers build concepts and make learning visible · Illustrations drawn for InquiryMind · Pictures from OpenMoji (openmoji.org), CC BY-SA 4.0 · Print in black and white on A4 · Challenge: ★ core ★★ deeper ★★★ stretch

Rocks, minerals and soils

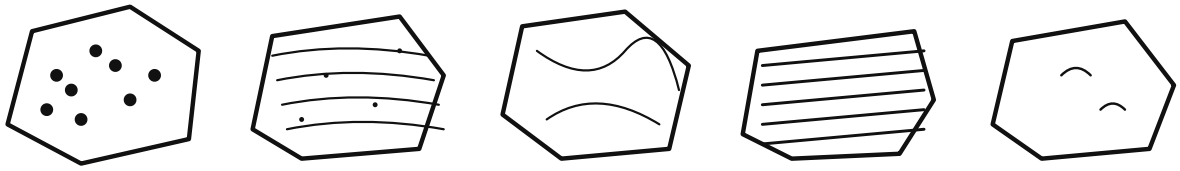
Name: Class: Date:

Learning goals

- Name some types of rock and their uses.
- Explain how rocks break down to make soil.
- Describe what soil is made of and how people use rocks, minerals and soil.

Rocks are everywhere: in hills, on beaches, under rivers and deep under your feet. Rocks are made of **minerals**. Over a very long time, rocks break into tiny pieces that become **soil**.

Types of rock



granite sandstone marble slate chalk

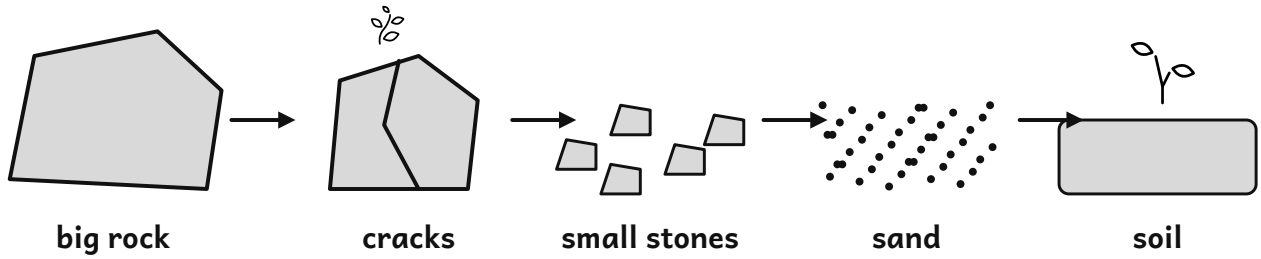
Five kinds of rock.

Rock	What it is like	We use it for...
granite	very hard, speckled	buildings, kitchen tops, statues
sandstone	made of sand grains stuck together	walls and buildings
marble	smooth, can be polished	floors, statues
slate	splits into thin flat sheets	roofs, writing boards
chalk	soft, white, crumbly	writing on boards

Minerals

Minerals are the natural materials rocks are made of. Salt is a mineral. Gold, iron and copper come from minerals in rocks. Shiny **gems** are minerals too.

From rock to soil



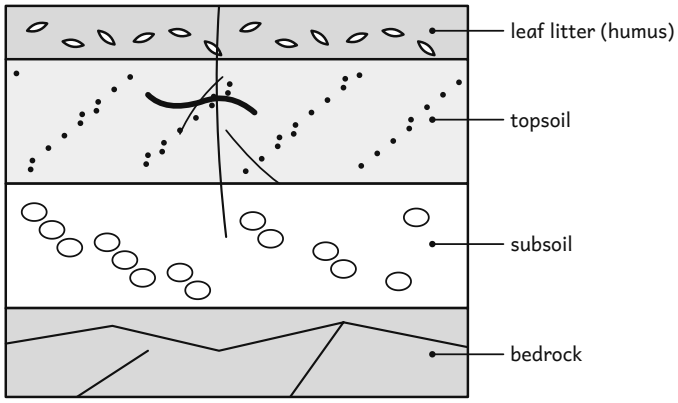
Rain, heat, ice and plant roots break rock into smaller and smaller pieces.

How rock becomes soil.

Rain, **heat**, **cold** and **plant roots** crack rocks. Rivers roll stones and rub them smooth and small. Over thousands of years rocks become **sand** and **soil**.

What is soil?

Soil is a **mixture** of tiny bits of rock (sand and clay), **dead plants and animals** (humus), **water** and **air**. It lies in **layers**.



Layers of soil.

How we use them

bricks from clay	stone for roads	salt	metals and gems	soil to grow food

Bangladesh connection

Bangladesh has very few rocks on the surface, so builders make **bricks** by baking clay soil in brick kilns. Stones for roads come from rivers in the north, such as near Sylhet.

Key words

rock	hard natural material from the Earth
mineral	a natural material that rocks are made of
soil	tiny bits of rock mixed with dead plants, water and air
humus	rotted plants and animals in soil

FACT CHECK · B2 · ROCKS AND SOILS

Rocks and their uses

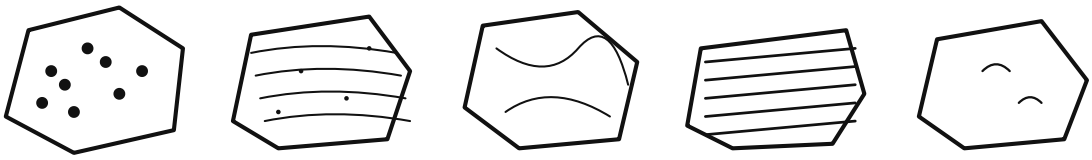
★☆☆ Grade 2

Name: Class: Date:

Learning goals

- Name rocks.
- Match rocks to uses.

1 Write the name of each rock. [5]



A B C D E

A	B	C	D	E

2 Draw a line to match the rock to its use. [4]

Draw a line to match each pair, or write the letter.

slate ●	● **A** roofs
chalk ●	● **B** floors and statues
marble ●	● **C** writing on boards
clay ●	● **D** bricks

3 What is soil made of? Tick. [4]

	In soil?
tiny bits of rock	
dead leaves (humus)	
water and air	
plastic	

Rocks change

★★☆
Grade 2

Name: Class: Date:

Learning goals

- Explain how rocks break down.

1 **Put the pictures in order to show how rock becomes soil.** [3]

- ☐ soil
- ☐ a big rock
- ☐ sand
- ☐ cracks in the rock
- ☐ small stones

2 **How can a tiny plant root break a rock?** [2]

.....

.....

3 **Why are pebbles in a river smooth and round?** [2]

.....

.....

INVESTIGATE · B2 · ROCKS AND SOILS

Which soil holds most water?

★★☆
Grade 2

Name: Class: Date:

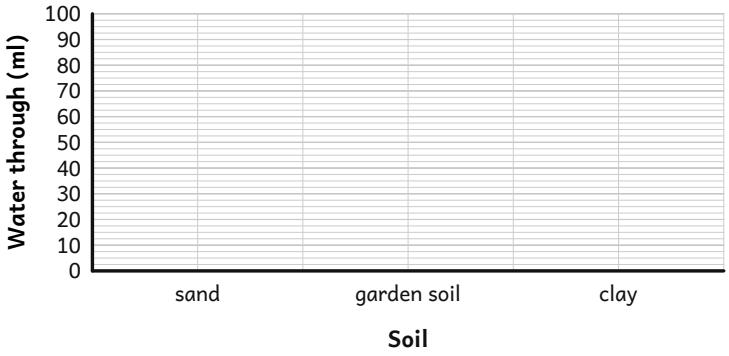
Learning goals

- Do a fair test.
- Measure water.

Class 2C put the same amount of sand, garden soil and clay into three funnels. They poured 100 ml of water on each and measured how much dripped through.

1 **Draw a bar chart of the water that dripped through.** [3]

Soil	sand	garden soil	clay
Water through (ml)	80	50	20



2 **Which soil let most water through?** [1]

3 **Which soil held most water?** [1]

4 **What did they keep the same?** [1]

REAL-WORLD CASE STUDY · B2 · ROCKS AND SOILS

The brick kiln

★★★
Grade 2

Name: Class: Date:

Learning goals

- Learn how soil is used to make bricks.
- Think about good and bad effects.

Science | Building | Environment

Near many towns in Bangladesh you can see tall chimneys. These are **brick kilns**. Workers dig up **clay soil**, mix it with water, shape it into blocks and bake them in a very hot fire. The soft clay becomes hard **bricks** for houses, schools and roads.

But brick kilns can cause problems: they use up good farm soil, and the smoke makes the air dirty. New kilns use cleaner fires and some builders use blocks made without burning.

Based on brick making in Bangladesh.

1 **Put the steps in order.** [4]

- ☐ hard bricks
- ☐ dig up clay
- ☐ bake in a fire
- ☐ mix with water
- ☐ shape into blocks

2 **Give one good thing and one problem about brick kilns.** [2]

.....

.....

Show what you know

★★★
Grade 2

Name: Class: Date:

Learning goals

- Use rock words.

1 Find the words.

H	B	T	O	K	M	I	S
A	G	S	N	R	O	C	K
C	M	C	L	C	L	A	Y
L	H	B	P	A	P	G	V
B	S	A	I	T	T	I	V
F	A	O	L	U	J	E	C
J	N	I	I	K	O	B	T
G	D	N	E	L	W	D	R

chalk clay rock sand slate soil

2 Which rock is soft and used for writing on boards? [1]

- ◯ granite
- ◯ chalk
- ◯ marble

3 Soil is made from tiny bits of... [1]

- ◯ plastic
- ◯ rock
- ◯ glass

4 Challenge. Draw the layers of soil and label two layers. [2]

Soil layers

How am I doing? Colour or tick a circle.

I can...	Not yet	Nearly	Got it!
name five rocks	○	○	○
say how rocks are used	○	○	○
explain how rock becomes soil	○	○	○
say what soil is made of	○	○	○
test which soil holds most water	○	○	○